



Reliability & Maintenance Decision Workbook

See It. Plan It. Do the Work. Learn From It.

A practical decision record for carrying asset evidence, professional judgment, uncertainty, execution context, and learning through the maintenance decision.

Asset / Equipment

Asset ID / Tag		Location / Facility	
System / Unit		Date Opened	
Work Order / Reference		Review Date	

What started this review?

Briefly describe the signal, observation, inspection result, recurring issue, operating change, or other reason the team is considering action.

Decision Team & Knowledge Contributions

Name	Role / Discipline	Contribution to the decision

Decision Owner / Technical Authority	
Work / Execution Owner	
Closeout / Learning Owner	

1. SEE IT - Establish the decision picture

Start with what is known. Keep established evidence distinct from interpretation and assumptions so the team can see the decision picture clearly.

Asset function - What does this asset need to do?

Established evidence - What can the team substantiate?

Examples may include monitoring trends, inspection findings, test results, operating history, maintenance records, photographs, measurements, or documented observations.

Interpretation - What do qualified people believe the evidence means?

Operating and field context - What matters around the asset right now?

Consider operating state, production needs, recent work, access, environment, resource constraints, related equipment, and relevant field experience.

Useful decision time - What opportunity has the evidence created?

The useful measure of prediction is the decision opportunity it creates.

IF	the evidence is still developing...
THEN	what additional information would materially improve the decision?

2. CHALLENGE IT - Make uncertainty visible

A strong decision can move forward with uncertainty. The important question is whether the uncertainty matters enough to change the action.

What is the team currently assuming?

What remains unresolved or materially uncertain?

Additional material observation or experience

Capture field knowledge or another credible interpretation that may affect the decision. Technical authority remains with the organization's qualified decision-makers.

What evidence would help distinguish between credible interpretations?

Would resolving this uncertainty change what we do?

■ YES - information may have decision value ■ NO - proceed with the current decision basis ■ REVIEW REQUIRED

IF	new evidence changes a material assumption or interpretation...
THEN	bring the qualified decision-makers back together and review the affected part of the decision.

3. PLAN IT - Choose the course and prepare the work

Record the practical choices, the reason for the selected course, and what must be ready to turn the decision into successful work.

Option	What would we do?	Timing / operating effect / key requirement
A		
B		
C		
D		

Selected course - What are we choosing to do?

Why does this choice make sense given what is known now?

Execution Readiness

Element	Ready?	What needs attention?
People / qualifications	■ Yes ■ Review	
Parts / materials	■ Yes ■ Review	
Tools / equipment	■ Yes ■ Review	
Procedure / work package	■ Yes ■ Review	
Access / isolation / permits	■ Yes ■ Review	
Operations / work window	■ Yes ■ Review	
Specialist / contractor support	■ Yes ■ Review	
Records / verification plan	■ Yes ■ Review	

What needs attention now to give this plan the best chance of delivering the expected result?

4. HAND IT OFF - Preserve the reasoning that matters

The next person may receive the work instruction. This page helps preserve the essential reasoning that should travel with it.

WHAT WE'RE DOING - Current decision

WHY - The reasoning the next person needs to understand

WHAT REMAINS OPEN - Material uncertainty, assumption, or alternate interpretation

Decision Condition

We are proceeding on the assumption that:	
IF:	
THEN review:	
Reconnect with / decision owner:	

What would be lost if the next person received only the work instruction?

Reasoning that must travel with the decision

5. DO THE WORK - Capture field reality

Execution creates evidence too. Give the people performing the work a clear place to record what became visible in the field.

What did the team actually find?

How did the actual condition compare with the pre-work understanding?

What became clear only during inspection, teardown, testing, or execution?

Did the field finding affect the current plan? If so, how was the decision reviewed?

IF	the field condition differs materially from what was expected...
THEN	pause the affected decision path and reconnect with the appropriate qualified owner under the organization's procedures.

Field contributor(s): _____ Date: _____

6. VERIFY IT - Compare expectation with reality

A good outcome and a good decision are related, but they are not identical. Compare what the team expected with what the work and subsequent operation actually showed.

Before the work	After the work
Expected condition / finding	Actual condition / finding
Expected work / resources	Actual work / resources
Expected result	Actual result / evidence

■ Result as expected ■ Result exceeded expectation ■ Result partly achieved ■ Additional follow-up identified

Given what was known at the time, was the reasoning appropriate? What supports that view?

What did reality reveal that was unavailable or unclear beforehand?

IF	the expected result is only partly achieved or new evidence appears...
THEN	identify what deserves review next: evidence, assumption, interpretation, execution condition, or maintenance strategy.

7. LEARN FROM IT - Improve the next decision

Close the loop by carrying useful learning forward. The goal is stronger future reasoning, not hindsight blame.

Which evidence proved most useful?

Which assumptions held? Which should be updated?

What should future teams recognize sooner?

What should change in future monitoring, planning, procedures, resources, or decision review?

Carry-forward areas

■ Monitoring ■ Inspection ■ Maintenance strategy ■ Planning ■ Scheduling ■ Parts / inventory
■ Procedure / work package ■ Training ■ Operating practice ■ Data / records ■ Contractor coordination ■ Other

What should the next team know that this team knows now?

Learning to carry forward

Follow-through owner

Review / completion date

Where the learning will be recorded

Decision Closeout

Evidence -> Interpretation -> Decision -> Reasoning Handoff -> Work -> Field Reality -> Learning

Final decision / disposition

Open actions and owners

Closeout summary

Decision Owner / Technical Authority		Date	
Execution Owner		Date	
Closeout / Learning Owner		Date	

Use & Boundaries

This workbook is a cross-functional decision companion. It supports structured reasoning and communication alongside an organization's established engineering, reliability, maintenance, operating, safety, work-control, permitting, regulatory, and management systems. Qualified personnel and organizational procedures remain the governing authority for technical and operational decisions.

Research Basis

Developed with reference to established asset-management, reliability, maintenance, condition-monitoring, work-control, human-factors, and decision-quality practices, including ISO 55001:2024, ISO 17359:2018, ISO 14224:2016, ISO 20815:2026, SAE JA1011:2024, GFMAM Maintenance Framework, and applicable regulatory/work-control principles.

Operational Trust

Preserve enough of the reasoning for the next person to understand what the organization believes, why it believes it, what remains uncertain, and what should cause the decision to change.